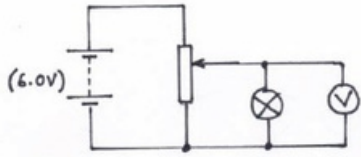
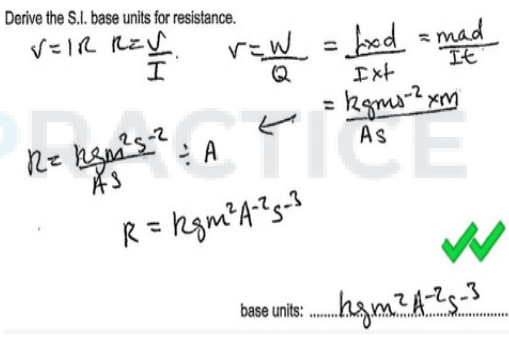


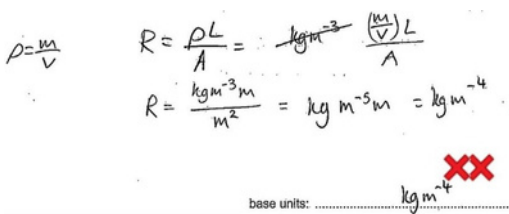
Question			Answer/Indicative content	Marks	Guidance
1			C	1	
			Total	1	
2			D	1	
			Total	1	
3	a		$\text{current} = \frac{0.060}{2.4}$ or $\text{current} = 0.025 \text{ (A)}$ $R = \frac{6.0 - 2.4}{0.025}$ $R = 140 \text{ (}\Omega\text{)}$	C1 C1 A1	Note answer to 3 sf is 144 Ω
	b		$I = Anev$ and $A = 2.0 \times 10^{-6} \text{ (m}^2\text{)}$ $0.025 = 2.0 \times 10^{-6} \times 1.4 \times 10^{25} \times 1.60 \times 10^{-19} \times v$ $v = 5.6 \times 10^{-3} \text{ (m s}^{-1}\text{)}$	C1 C1 A1	Allow any subject Possible ecf
	c		The current is constant, therefore $v \propto n^{-1}$. The mean drift velocity is therefore smaller. Total	M1 A1 8	
4			$Q = It$ and $e = 1.6 \times 10^{-19} \text{ (C)}$ number of electrons = $0.24 \times 10^{-6} \times 5.0 / 1.6 \times 10^{-19}$ number of electrons = 7.5×10^{12} Total	C1 C1 A1	
			B	3	
5			Total	1	
			Correct circuit with a battery, potential divider, lamp and voltmeter.	1	
6		i		B1	

Question			Answer/Indicative content	Marks	Guidance
		i	Correct symbols used for all components.	B1	Allow: A cell symbol for a battery
		ii	Description: The temperature of the filament increases. (AW)	B1	Allow 'when cold the resistance is small'
		ii	The resistance of the lamp increases	M1	
		ii	from a non-zero value of resistance.	A1	
		ii	Explanation: Resistance increases because electrons/charge carriers make frequent collisions with ions. (AW) ($P = VI$) current in X is 3 times the current	B1	
		iii	in Y Or area of X is 4 times smaller than area of Y	C1	Allow other correct methods.
		iii	$I = Anev$ and ratio = $\frac{3}{0.25}$	C1	
		iii	ratio = 12	A1	
			Total	9	

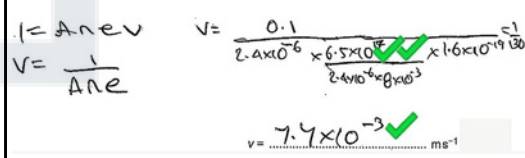
Question			Answer/Indicative content	Marks	Guidance
7			B	1	Examiner's Comment All questions showed a positive discrimination, and the less able candidates could access the easier questions. The questions in Section A require careful reading and execution. Underlining or circling key information may help. Candidates are reminded not to use highlighter pens for this purpose. There is ample space for jotting down ideas and key equation, but it is best to do calculations on calculators to save time. Questions 1, 3, 10 and 14 proved to be particularly straightforward, allowing most of the candidates to demonstrate their knowledge and understanding of physics. At the other end of the scale, Questions 5, 9, and 15 proved to be more challenging. • Question 5 was on the superposition of waves and the relationship $\text{intensity} \propto \text{amplitude}^2$. The amplitude of the resultant wave is $0.4a$ and therefore the intensity of resultant wave must be $0.16I$. The most popular distractors were B and C. Less than half of the candidates got the correct answer of A. • Question 9 was about doubling the separation between two oppositely charged parallel plates. The only correct statement is D. Electric field strength is p.d. divided by the distance between the plates. Since both quantities double, the electric field strength, must remain the same. • Question 15 was about refraction and the equation $n_1 \sin i = n_2 \sin r = \text{constant}$ at the boundary between two materials. The ratio $n_1/n_2 = \sin 80^\circ / \sin 90^\circ = 0.98$; the correct is B. The most popular distractor was C, which was the inverse of the correct answer.
			Total	1	
8			C	1	
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
9		i	$I = nAev;$ $v = 60 \times 10^{-3} / 1.2 \times 10^3 \times 1.6 \times 10^{-19}$ $\times 5.0 \times 0.2 \times 10^{-6}$ $v = 3.1 \text{ (m s}^{-1}\text{)}$	C1 C1 A1	allow any subject
		ii	$V = 80 \times 10^{-3} \times 3.1 \times 5.0 \times 10^3$ $= 1.2 \times 10^{-3} \text{ (V)}$	A1	ecf (b)(i); allow 1.2 mV; $1.3 \times 10^{-3} \text{ (V)}$ Examiner's Comments This exercise of choosing a formula, substituting values in correct units and evaluating was done well with about three quarters of the candidates gaining full marks.
			Total	4	
10		i	$2.76 - 2.3 = 0.46 \text{ eV}$ (so only 0.5% of energy/AW)	B1	allow $2.8 - 2.3 = 0.5 \text{ eV}$ and $3.0 - 2.3 = 0.7 \text{ eV}$ possible ecf from (b)
		ii	$n = 2000 \times 49 (= 5.24 \times 10^8)$ $Q = ne = 8.4 \times 10^{-11} \text{ (C)}$ $I = 8.4 \times 10^{-11} / 2.5 \times 10^{-9}$ average current = 0.034 (A)	C1 C1 A1	allow ecf for wrong n allow 34 m(A); answer is $1.7 \times 10^{-5} \text{ A}$ if 2000 omitted (2/3) Examiner's Comments Almost all of the candidates attempted this last section of the paper with some success. In part (i) most candidates showed that they understood the theory behind the question and subtracted the appropriate two numbers from part (b) to gain the mark. Part (ii) was done well with a significant number obtaining the correct answer. Another large group forgot that 2000 electrons were released and performed the calculation for only a single electron being multiplied up and so forfeited the final mark.
			Total	4	
11			A	1	
			Total	1	

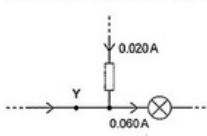
Question		Answer/Indicative content	Marks	Guidance
12	a	$(R = \frac{V}{I} = \frac{W}{QI}; Q = It)$ charge → A s or energy → kg m ² s ⁻² × m or kg m ² s ⁻² (base units) kg m ² A ⁻² s ⁻³	C1 A1	Allow other correct methods Allow Q or C or coulomb for 'charge'; E or W or joule or J or work done for 'energy' Allow 1 mark for J s⁻¹ A⁻² Allow $\frac{\text{kg m}^2}{\text{A}^2 \text{ s}^3}$ or kg m² /(A² s³) Not kg m² / A² / s³ or kg m² / s³ / A² Examiner's Comments This was a challenging question, which provided the ideal opportunity for top-end candidates to use a variety of methods to get the correct S.I. base units of kg m² A⁻² s⁻³ for resistance. A significant number of candidates secured 1 mark for a partial answer with either charge → A s, or energy → kg m² s⁻². The rules for exponents were a bit perplexing for the low-scoring candidates. Many also misunderstood S.I. units. Exemplar 4  This exemplar illustrates a flawless answer from a top-end candidate. The equations are clear to see and follow. The units of each physical quantities are clearly identified and the appropriate S.I. units for the quantities have been

Question			Answer/Indicative content	Marks	Guidance
					successfully manipulated to give the correct answer. Compare this with the exemplar below which illustrates a common misconception. Exemplar 5  This exemplar illustrates a common error made by some candidates across the ability spectrum. ? The resistivity ρ in the equation for resistance has been mistaken for density (which unfortunately has the same label). There can be no credit for wrong physics. It is vital to know your equations. Key: ? Misconception

Question			Answer/Indicative content	Marks	Guidance
	b	i	$(R =) \frac{6.0}{0.150}$ $R = 40 \Omega$	M1 A0	Allow any correct value of V (± 0.1 V) divided by the correct value of I (± 10 mA) from the straight line for R <u>Examiner's Comments</u> The majority of the candidates scored 1 mark here for clearly using the graph to show the resistance of R to be 40Ω. Most used a data point from the straight line. A significant number also used the idea that the gradient of the straight line is equal to the inverse of the resistance. However, candidates are reminded that resistance is equal potential difference divided by current, but in this context of a straight line through the origin, determining resistance from the gradient was allowed. Of course, determining the gradient of a curve is simply incorrect physics for determining resistance.
		ii	$(V_L =) 1.4$ (V) or $(V_R =) 4.0$ (V) or $(R_T =) 6.0/0.1$ (Ω) $(V_{\text{terminal}} =) 5.4$ (V) or $(V_r =) 0.6$ (V) or $(r =) 60 - 54$ (Ω) $r = 6.0$ (Ω)	C1 C1 A1	Allow full credit for other correct methods Possible ECF from (i) Allow ± 0.1 V for the value of p.d. from the graph Note getting to this stage will also secure the first C1 mark Allow 1 SF answer here without any SF penalt <u>Examiner's Comments</u> This was a discriminating question with many of the top-end candidates effortlessly getting the correct answer of 6.0Ω for the internal resistance r. The most common error was omitting the resistance of the filament lamp in the calculation. This gave an incorrect value of 20Ω for the internal resistance. Candidates doing this still managed to pick up 1 mark for the total resistance of 60Ω.

Question			Answer/Indicative content	Marks	Guidance
		iii	$\rho = \frac{40 \times 2.4 \times 10^{-6}}{8.0 \times 10^{-3}}$ (Any subject) $\rho = 0.012 \text{ } (\Omega \text{ m})$	C1 A1	Allow ECF Allow 1 mark for either 0.018 for using 60 Ω, 0.016(2) for using 54 Ω or for 0.0018 for 6.0 Ω Examiner's Comments The success in this question depended on understanding the term n in the equation $I = Anev$ given in the Data, Formulae and Relationship booklet. A significant number of candidates took n to be the total number of charge carriers within the volume of R, instead of the number of charge carriers per unit volume (number density). Those who appreciated this had no problems coping with prefixes and powers of ten. The correct answer was $7.7 \times 10^{-3} \text{ m s}^{-1}$. Using 6.5×10^{17} for the number density, gave an answer of $4.0 \times 10^5 \text{ m s}^{-1}$; examiners credited 1 mark for this incorrect answer, mainly for the manipulating and using the equation $I = Anev$. Exemplar 6  This exemplar illustrates a perfect answer from a C-grade candidate. The equation has been rearranged correctly and the substitution is all correct and easy to follow. The number density n has not been calculated separately – it forms an integral part of the whole calculation. The one big benefit of this is that you do not end up with rounding errors. A decent technique demonstrated here. All correct for 3 marks.

		candida because coulom
	1	

Question	Answer/Indicative content	Marks	Guidance
14	B	1	<u>Examiner's Comments</u> This was not an easy question, but most candidates did extremely well in this multi-step calculation. The directions of the currents are important. The current at Y and the current in the resistor, must add up to 0.060 A. The charge passing point Y in a time of 10 s can be calculated using $\Delta Q = I\Delta t$ and finally the number of electrons can be determined by dividing the charge passing through Y by the elementary charge e. Therefore $\text{number of electrons} = \frac{(0.060 - 0.020) \times 10}{1.60 \times 10^{-19}} = 2.50 \times 10^{18}$ The key, correct answer, is B. The most popular distractor was C, in which a current of 0.060 A is assumed; the rest of the distractors were equally favourable. A was the answer for a current of 0.020 A and D was the answer for a current of 0.080 A. It is worth mentioning that most of the candidates just wrote down the correct answer in the box, without any calculations. On the surface, this looks reckless, but it is an excellent strategy if the numbers are being punched into the calculator correctly. The exemplar 1 below shows a typical correct answer, with numbers jotted down for visual help. Exemplar 1 Part of an electric circuit is shown below.  The direction of all the currents and the magnitude of two currents are shown. How many electrons pass through the point Y in 10 s? A 1.25×10^{18} B 2.50×10^{18} C 3.75×10^{18} D 5.00×10^{18} Your answer B [1] The candidate has the correct current at Y.

Question			Answer/Indicative content	Marks	Guidance
					Without the 0.040 A, the answer would have led to one of the distractors. It is worth pointing out that the final step of the calculation must have been done directly on the calculator. An excellent time-saving approach.
			Total	1	
15			B	1	<u>Examiner's Comments</u> This was a question on combining together three important expression in the topic of electricity; $V = IR$, $R = \rho L/A$ and $I = Anev$. On top of this, there was the additional information that P and Q were in parallel and hence the potential difference across each wire was the same. The mean drift velocity v of the electrons is given by the expression $v = \frac{V}{ne\rho L} \propto \frac{1}{L}$. The cross-sectional area A, and hence the diameter d of the wire has no effect on v. The relationship above implies that for wire Q, $v = \frac{1}{3} \times 0.60 = 0.20 \text{ mm s}^{-1}$. The correct answer is B. All the distractors were equally popular. About a third of the candidates, mostly from the very top end of the ability range, were successful in this very demanding question.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
16	a	i	$R = \frac{150}{1.5^2}$ 67 Ω	C1 A1	Allow $V = \frac{150}{1.5} = 100 \text{ V}$ <u>and</u> $R = \frac{100}{1.5}$
		ii	$Q = 1.5 \times 5.0 \times 60 \times 60 \text{ or } 27000$ $N = \frac{1.5 \times 5.0 \times 60 \times 60}{1.6 \times 10^{-19}} = 1.7 \times 10^{23}$	C1 A1	Note use of 150 (W) does not score 1.7 $\times$ 1025 1.68 $\times$ 1023 4.7 $\times$ 1019 scores one mark Not 1.7 $\times$ 1025 (uses 150 W)
		iii	$v = \frac{1.5}{7.9 \times 10^{28} \times 4.1 \times 10^{-9} \times 1.6 \times 10^{-19}}$ 0.029 (m s ⁻¹)	C1 A1	
	b		Silicon will have a smaller number density, ORA Silicon will have a larger resistivity, ORA	B1 B1	Allow semiconductor for silicon; metal for nichrome <u>Examiner's Comments</u> High achieving candidates found this question straightforward. Some candidates on (a)(iii) used N instead of n . Part (b) caused the most difficulty with candidates either using 150 W rather than 0.150 kW or changing the time to seconds. ? Misconception The worst acceptable line is either the steepest line that passes within all the error bars or the shallowest error line that passes within all the error bars.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
17		i	$12\,000 = \frac{Q}{4\pi\epsilon_0 r^2}$ $12\,000 = \frac{Q}{4\pi\epsilon_0 \times 0.19^2}$ $Q = 2.5(4) \times 10^{-7} \text{ (C)}$	C1 C1 A0	Allow $E = (V/d) = 6.316 \times 10^4$ and $E = 6.316 \times 10^4 = \frac{Q}{4\pi\epsilon_0 \times 0.19^2}$ C1 C1
		ii	$1t = 78 \times 3600$ $(I =) \frac{2.5 \times 10^{-7}}{78 \times 3600}$ $I = 8.9 \times 10^{-13} \text{ (A)}$ 2 $(R =) \frac{6000}{9.0 \times 10^{-13}} \text{ or } 6.7 \times 10^{15} \text{ (}\Omega\text{) or } V =$ $IR \text{ and } R = \frac{\rho L}{A}$ $\frac{6000}{9.0 \times 10^{-13}} = \frac{\rho \times 0.38}{1.1 \times 10^{-4}}$ $\rho = 1.9 \times 10^{12} \text{ (}\Omega \text{ m)}$	C1 C1 A0 C1 C1 A1	There is no ECF from (b)(i) Note 2.54×10^{-7} gives an answer $9.0 \times 10^{-13} \text{ A}$ There is no ECF from (b)(ii)1 Take 12000 V as TE for this C1 mark, then ECF Note $8.9 \times 10^{-13} \text{ (A)}$ gives an answer $2.0 \times 10^{12} \text{ (}\Omega \text{ m)}$
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
18	a	i	$F = QE = QV / d$ or $E = 5(.0) \times 10^4 \text{ (Vm}^{-1}\text{)}$ $F = 9.0 \times 10^{-9} \times 4000 / 8.0 \times 10^{-2} (= 4.5 \times 10^{-4} \text{ N})$	C1 A1	$F = 5.0 \times 10^4 \times 9.0 \times 10^{-9}$ Examiner's Comments Many lower ability candidates did not appreciate the uniform nature of the electric field between the plates and attempted to use Coulomb's Law.
		ii	weight; arrow vertically downwards tension; arrow upwards in direction of string electric (force); arrow horizontally to the right (not along dotted line) —	B1 x 2	All correct, 2 marks; 2 correct, 1 mark 1 mark maximum if more than 3 arrows are drawn Ignore position of arrows Allow W or 0.030(N) (not gravity or g) Allow T Allow F or E or $4.5 \times 10^{-4} \text{ (N)}$ or electrostatic Ignore repulsion or attraction Not electric field / electric field strength / electromagnetic Examiner's Comments Most candidates scored a mark for — showing the weight and tension forces accurately. Only a small proportion labelled the electric force arrow correctly and drew it as clearly perpendicular to the plates.  Afl Do not use the word 'gravity' in place of 'weight'

Question			Answer/Indicative content	Marks	Guidance
		iii	$W x = F l$ $0.03 x$ $= 4.5 \times 10^{-4} \times 120 \text{ or } = 4.5 \times 10^4 \times 1.2$ $x = 1.8 \text{ cm or } x = 0.018 \text{ m}$	M1 M1 A0	Allow any valid alternative approach e.g. M1 deflection angle $\theta = 1^\circ$ $M1 x = 120 \sin \theta$ 1 mark for each side of the equation <u>Examiner's Comments</u> Although most candidates knew the principle of moments, many were unable to apply it correctly in this situation. More practice at this sort of question is recommended.
	b		Electric force/field (strength) increases Ball deflected further from vertical / moves to the right / touches negative plate Ball acquires the charge of the (negative) plate when it touches (Oscillates because) constantly repelled from (oppositely) charged plate	B1 B1 B1 B1	Must be clear which force is increasing Must have the idea of a repeating cycle <u>Examiner's Comments</u> The purpose of this question was to challenge the candidates to use their knowledge of electric fields in a novel practical situation. The word 'oscillate' confused many candidates, who tried to explain why the ball would perform simple harmonic motion.
	c		$I = Qf \text{ or } Q = It$ $f = 3.2 \times 10^{-8} / 9.0 \times 10^{-9} = 3.6 \text{ (Hz)}$	C1 A1	
			Total	12	

Question			Answer/Indicative content	Marks	Guidance
19	a		(initial charge) $Q = EC_0$ or (Q conserved so final) $Q = V(C + C_0)$ (as capacitors are in parallel) so $EC_0 = V(C + C_0)$ (and hence $V = C_0 E / (C + C_0)$)	M1 A1	At least one correct expression for Q for first mark The two correct expressions equated for the second mark Examiner's Comments Some candidates obtained $Q = EC_0$ by applying the definition of capacitance at A, but then did not realise that charge would be conserved on switching from A to B. Some chose the wrong formula for capacitors in parallel or attempted to use the potential divider equation. Mark is for rearrangement into linear
	b	i	$1/V = 1/E + C/EC_0$ (and compare to $y = c + mx$)	B1	equation Examiner's Comments Some candidates correctly took the reciprocal of both sides of the given equation but were then unable to show a rearrangement into the standard linear form. A common difficulty was an inability to expand the bracket in $\frac{1}{E} \times \frac{(C + C_0)}{C_0}$ to give $\frac{C}{EC_0} + \frac{C_0}{EC_0}$

Question			Answer/Indicative content	Marks	Guidance
		ii	$\frac{1}{EC_0} = 51 = \frac{1}{(9.1 \times 10^{-6})}$ giving $C_0 = \frac{1}{(51 \times 9.1)} \text{ F}$ $C_0 = 2.2 \text{ (mF)}$	B1 B1	$C_0 = 2.1547 \times 10^{-6}$ Answer must be correct, rounded correctly and given in mF Candidate's answer must be given to 2 SF Examiner's Comments Some candidates gave their response to 2 d.p. instead of to 2 s.f. as required.

Question			Answer/Indicative content	Marks	Guidance
		iii	(at least) one correct worst fit line drawn gradient calculated correctly using a large triangle $\text{uncertainty} = C_0 - 1/(\text{wfl gradient} \times 9.1)$ uncertainty given is to the same number of decimal places as C_0	B1 B1 B1 B1	Top and bottom points chosen must be from opposite extremes of uncertainty limits, accurate to within half a small square $\Delta x \geq 1.5 \times 10^{-3}$; expect 59 ± 1 or 44 ± 1 (or 0.059 or 0.044); allow ECF from poorly drawn line; readings must be accurate to within half a small square ECF from b(ii); expect uncertainty of up to 0.4(mF) ECF from b(ii) If no value for C_0 given in b(ii), allow any answer given to 1dp Examiner's Comments Most candidates gained the mark for using a large triangle (spanning more than 1.5 on the x-axis) to determine the gradient of the worst-fit line. Lower ability candidates were unable to gain credit for finding the gradient of their line because they read the scales on the axes incorrectly. Candidates should take a ruler into the examination and be careful about the positioning of the ruler for drawing a worst-fit straight line. A worst-fit line should join opposite extremes of uncertainty limits and pass between all the uncertainty limits. The Practical Skills Handbook is helpful on this topic. Several candidates performed the unnecessary step of calculating the fractional (or percentage) uncertainty instead of using $\Delta C_0 = \pm C_0 \text{ best} - C_0 \text{ worst} $ directly. Allow and so the experiment takes longer
	c		Only effect is to slow the charging and / or discharging (of capacitor(s)) <u>and so</u> the final charges are unchanged / the values for V are unchanged / the graph is unchanged / the gradient is unchanged / there is no effect on the experiment (results)	B1	
			Total	10	

Question	Answer/Indicative content	Marks	Guidance
20	Level 3 (5 – 6 marks) Clear expansion of three statements <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is clear, relevant and substantiated.</i> Level 2 (3 – 4 marks) Clear expansion of two statements or Limited attempt at all three <i>There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.</i> Level 1 (1 – 2 marks) Limited attempt at one or two statements <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 marks <i>No response or no response worthy of credit.</i>	B1 x 6	Use level of response annotations in RM Assessor, e.g. L2 for 4 marks, L2 ~ for 3 marks, etc. Indicative scientific points may include: statement 1 • fusion reactions are occurring • which change H into He • and mass is lost which releases energy • energy released = $c^2\Delta m$ • Δm per second = luminosity / c^2 statement 2 • average k.e. of each proton is $\frac{3}{2}kT$ • high T means protons are travelling at high speed • so fast enough to overcome repulsive forces • and get close enough to fuse • p.e. = $e^2/4\pi\epsilon_0 r$ so T must be high enough for $\frac{3}{2}kT > e^2/4\pi\epsilon_0 r$ • r is approximately 3fm statement 3 • k.e. $\propto T$ so average energy at 107 K is only one thousandth of the average energy at 1010 K when protons might fuse • but M-B distribution applies so at the high energy end there will be a few p with enough energy • quantum tunnelling across potential barrier is possible • small probability of many favourable collisions to boost energy of p • 4 p must fuse to produce He; it is complicated process making probability of fusion much less • number of p in Sun is so huge that, • even with such a small probability, 4×10^{57} kg of p still interact s⁻¹ • a larger probability means lifetime of the Sun would be shorter

Question			Answer/Indicative content	Marks	Guidance
					<u>Examiner's Comments</u> This was one of the two LoR questions. It required understanding of fusion, mass-energy equivalence, the Maxwell-Boltzmann distribution, and the relationship between mean kinetic energy and temperature for particles in an ideal gas. Responses to the following questions were being sought: 1. Why is the Sun losing mass? 2. Why is an extremely high temperature needed for fusion in stars? 3. Why does fusion occur in the Sun even though its temperature is 1,000 times less than that required by theory? Two dissimilar responses could score comparable marks if the criteria set out in the answer section of the marking scheme were met. Level 3 responses gave a clear answer to all three of the questions, whereas Level 2 responses generally had clear answers to only two. In Level 1, limited answers to only one or two of the above questions were given.
			Total the current (induced in the aerial) is alternating (5×10^8 times per second) (so the meter would register zero) / AW	6	
21			or the diode (half-)rectifies the current / changes the current (from a.c.) to d.c. / AW	B1	Allow 'a diode only lets current pass through in one direction' AW <u>Examiner's Comments</u> <u>Allowing a mark for the diode only letting current pass in one direction</u> enabled many candidates to score this mark. There was little mention of alternating current among the responses.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
22			$Q = 79e$ and $q = 2e$	C1	Apply ECF for wrong charge(s), e.g. Q and / or $q = e$, or $Q = 79$ and / or $q = 2$, etc Examiner's Comments The most common error here was to use incorrect values for the charges on the two ions. Even so, most candidates were able to gain most of the marks with ECF.
			$F = (1/4\pi\epsilon_0)Qq/r^2$	C1	
			$= 79 \times 2 \times (1.6 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (6.8 \times 10^{-14})^2]$	C1	
			$= 7.9 \text{ (N)}$	A1	
			Total	4	
23			C	1	
			Total	1	



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